

Research papers

Assessing the 3E performance of multiple energy supply scenarios based on photovoltaic, wind turbine, battery and hydrogen systems

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ABSTRACT

Supplying buildings with cost-effective, eco-friendly and reliable energy by hybridizing sustainable renewable energy systems (RESs) and energy storage methods is becoming increasingly interesting. This study investigates the energy, environmental and economic (3E) effectiveness of three energy supply scenarios (ESSs) designed to meet the electricity and thermal needs of a typical residential building under the Mediterranean climate of Morocco. The first scenario aims to directly supply the energy produced by two RESs, namely photovoltaic and wind systems, to the building and inject the surplus electricity into the grid. In contrast, in the second scenario, the excess solar and wind energy is stored in batteries, while in the third one a proton exchange membrane electrolyzer is used to produce hydrogen and store it for later use through fuel cells. To examine the performance of these scenarios, numerous software, such as EnergyPlus and PVsyst, as well as numerical models are used. The findings indicate that the second and third scenarios can achieve an energy coverage ratio of up to 100 %, whilst the first scenario reaches lower ratio of 52.56 %. Environmentally, the first scenario exhibits the best outcomes with a carbon payback period of one year and a reduction in carbon emissions of 18.21 tCO_{2e}/year. On the other side, the carbon payback period and carbon emissions reduction are 1.66 years and 15.13 tCO_{2e}/year for the second scenario, and 1.14 years and 14.82 tCO_{2e}/year for the third scenario. Economically, the first scenario reveals the best economic findings, followed by the second one, while the third scenario is not economically feasible. The obtained simple payback period values for the first, second and third scenarios are 5.80, 7.89 and 16.30 years, whereas their levelized cost of energy shares are 0.05, 0.09 and 0.21 \$/kWh, sequentially. The outcomes of this research significantly contribute to the field by providing crucial information for building owners and professionals to help them in discerning the most efficient ESSs in the 3E prospects, and hence improving their decision-making and implementation processes.

1. Introduction

1.1. Background

The increased exploitation of fossil fuels has raised the energy usage's environmental burden. This issue is particularly pressing in the building sector, accounting for around 39 % of global energy-related carbon emissions [1]. Subsequently, shifting towards clean and sustainable energy supply chains becomes an urgent need to mitigate the negative energy, social, economic and environmental impacts of buildings [2]. In this vein, the integration of renewable-based systems instead of fossil fuel-based ones for generating electricity, and providing the required heating and cooling loads for buildings is crucial [3]. With the increasing development and significant cost reductions of renewable

energy systems (RESs), especially solar and wind-based ones, their use for improving the sustainability of buildings and ensuring clean energy supply is becoming increasingly important [4,5]. The variant possibilities to store the surplus energy produced by RESs are also one of their primary advantages. Currently, two main storage methods can be integrated with RESs and used in buildings: conventional electrochemical energy storage and hydrogen storage.

Conventional electrochemical energy storage devices, like lithium-ion batteries, provide high energy efficiency and cost-effectiveness but they suffer from limitations like limited capacity, environmental concerns and shortened lifespan [6]. In contrast, hydrogen is emerging as a promising and viable solution for energy storage. Its production can be carried out through diverse methods, including renewable and fossil-based routes [7]. Regarding renewable-based processes, water electrolysis is the most mature and eco-friendly technique for producing

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Nomenclature		G_T	solar radiation received by the PV module, kWh/m ²
Abbreviations		h	number of hours, hours
CAPEX	Capital Expenditure, \$	I_t	investments during the t-th year, \$
CER	Carbon Emissions Reduction, tCO _{2e} /year	LHV	lower heating value of hydrogen, kWh/kg
CPBP	Carbon Payback Period, years	m	mass, kg
DPP	Discounted Payback Period, years	N	lifetime, years
ECR	Energy Coverage Ratio, %	P	power, kW
ESS	Energy Supply Scenario	Q	thermal energy, kWh
HVAC	Heating, Ventilation and Air Conditioning	q_{net}	net heat transfer rate into the water tank, kW
LCOE	Levelized Cost of Energy, \$/kWh	SoC_{min}	minimum state of charge, %
LCOH	Levelized Cost of Hydrogen, \$/kgH ₂	S_t	savings during the t-th year, \$
OPEX	Operating Expenditure, \$	T	temperature, K
PEM	Proton Exchange Membrane	t	time, s
PV	Photovoltaic	V	wind speed, m/s
RES	Renewable Energy System	V_B	nominal voltage of battery, V
SIR	Savings-to-Investment Ratio	Greek symbols	
SPP	Simple Payback Period, years	β	temperature coefficient, K ⁻¹
WT	Wind Turbine	η	efficiency, %
3E	Energy, Environmental and Economic	ρ	air density, kg/m ³
Symbols		Δt	time step, s
A_{PV}	surface area of PV panel, m ²	Subscripts	
A_r	swept area of rotor, m ²	B	battery
B_{aut}	battery autonomy, hours	$cell$	PV cell
C_B	battery capacity, Ah	ch	charged
CEF	carbon emission factor, kgCO _{2e} /kWh	$disch$	discharged
CF	capacity factor of a WT, %	ele	electrolyzer
C_F	annual cash flow, \$	FC	fuel cell
C_p	performance coefficient	gen	generated loads
c_p	specific heat, J/(kg.K)	H_2	hydrogen
d	discount rate, %	$invert$	inverter
E	electricity, kWh	req	requested loads
EC	embodied carbon, kgCO _{2e}	SC	scenario
E_t	generated/stored energy during the t-th year, kWh	ST	storage tank
f_{active}	fraction of active PV panel area to the total PV panel surface area, %	STC	standard test conditions

green hydrogen [8] that can be stored, and/or transformed into electricity and heat using fuel cells [9]. This heat can be harnessed in diverse applications in buildings, mainly for space heating, water heating and cooking. Overall, compared to energy supply scenarios (ESSs) based on batteries, hydrogen-based ones offer additional benefits, including higher energy density, lower carbon footprint and extended lifespan [10]. However, the elevated energy efficiency, cost-effectiveness and maturity of batteries make them a preferred energy storage option for building owners over hydrogen storage [11].

1.2. Literature review

Recently, many countries around the world have undertaken efforts to decarbonize their final energy consumption, especially those with abundant renewable energy resources, such as Morocco. In this context, the Moroccan's authorities have launched numerous initiatives to exploit its exceptional potential in renewable energy, using mainly photovoltaic (PV) and wind turbine (WT) systems [12,13]. Nonetheless, the wider adoption of these systems still faces major barriers, particularly concerning their intermittent generation and the energy, economic and environmental (3E) efficacy of energy storage systems.

In the literature, a dearth of studies examining the 3E performance of ESSs employing RESs, and batteries or hydrogen storage in Morocco is noticeable. For example, Ourya et al. [14] examined the economic profits of utilizing a hybrid PV-WT system powering an electrolyzer for

hydrogen production in five Moroccan cities. The findings reported that the levelized cost of hydrogen (LCOH) could be as low as 2.54 \$/kg. Similarly, Khouya [15] evaluated the costs of energy and hydrogen associated with hydrogen generation employing an alkaline water electrolyzer supplied by electricity generated from wind farms and concentrated PV thermal plants in the Mediterranean climate of Tangier. The findings indicated that the expenditures related to energy and hydrogen produced using WTs were marginally elevated compared to those harnessing concentrated PV plants. Ennemiri et al. [16] assessed the economic and environmental impacts of integrating a hybrid energy plant, combining PV, biogas and battery systems, aimed to supply electricity to a commercial platform located in Berkane. They revealed that using the hybrid system resulted in an electricity cost of 0.28 \$/kWh and a decrease in CO₂ emissions of almost 40 % in comparison to an ESS based on a standalone biogas system. These works underscore the crucial role of incorporating RESs and energy storage technologies in enhancing the sustainability of building energy supply networks in Morocco.

On a global scale, many researchers have studied the performance of integrating different building ESSs, including both RESs and energy storage methods. For example, Jung et al. [17] analyzed the performance of net zero energy buildings supplied by two distinct energy sources: electricity and hydrogen. The results revealed that the electricity-based scenario outperformed the hydrogen-based one as concerns both energy efficiency and environmental effect. Jin et al. [18]

demonstrated that the use of batteries is an effective way for energy storage, but the high capital expenditure (CAPEX) is their main drawbacks. Moreover, they reported that the incorporation of hydrogen in buildings is not economically viable, which can mainly be attributed to the high CAPEXs. Likewise, Wang et al. [19] indicated that hydrogen storage was more eco-friendly than electrochemical storage. However, the implementation of the hydrogen storage method proved to be more expensive, reducing its cost-effectiveness. On the other hand, Assareh et al. [20] evaluated the performance of a combined cooling, heating and power system comprising various RESs and energy storage units. This system includes PV thermal panels, WTs, an electrolyzer, fuel cells and a heat pump, while the energy storage package consists of a hydrogen storage tank and a battery system. They concluded that the integration of both hydrogen storage tank and batteries is essential to manage the stored energy and reduce the unpredictability in supply and demand imbalances. Xu and Chen [21] investigated the effectiveness of integrating a rooftop PV system with hybrid electric, hydrogen and thermal energy storage in a residential building. The explored system exhibited excellent economic benefits, achieving a low levelized cost of energy (LCOE) of 0.10 \$/kWh. Actually, the economic profits of this system were greater when it was connected to the grid compared to the stand-alone option. Environmentally, they noted that this system is more eco-friendly when supplied by PV collectors rather than the local grid. Therefore, a comprehensive life cycle assessment and comparison of the 3E performance of various ESSs is needed in order to gain deeper insights into their overall efficiency and limitations.

1.3. Research gaps and key contributions

Based on the above works and an extensive review of the literature, it is evident that despite the considerable research devoted to studying the feasibility of some ESSs, significant gaps remain unaddressed. Primarily, there is a lack of detailed life cycle assessments of multiple ESSs based on RESs, energy storage systems, and the grid. Previous studies have predominantly concentrated on the energy and/or economic aspects of ESSs, neglecting other crucial prospects, such as their environmental performance. Moreover, the majority of prior research focused solely on the provision and storage of a single type of energy, such as electricity or heat, without considering the potential synergies of a more holistic approach. Additionally, most studies have been conducted in developed countries, distinguished by specific climatic, environmental, and economic contexts, thus limiting the generalizability of findings to broader global scenarios. These gaps highlight the need for more comprehensive research to examine and compare the sustainability of different ESSs under various conditions. In this respect, the current study aims to reach the following objectives:

- Perform detailed life cycle assessments of the energy performance, environmental friendliness and cost-effectiveness of three ESSs, taking into account their energy losses, embodied carbon and life cycle costs. These ESSs are designed to store and provide electricity, heating, cooling and hot water for a typical residential building in the Mediterranean climate of Morocco.
- Assess the effectiveness of the proposed ESSs utilizing a combination of software tools, including EnergyPlus and PVsyst, along with numerical modeling techniques.
- Analyze the results using key metrics in the 3E perspectives, namely energy production, energy coverage ratio (ECR), carbon payback period (CPBP), carbon emissions reduction (CER), simple payback period (SPP), discounted payback period (DPP), savings-to-investment ratio (SIR) and LCOE.

Reaching these objectives would offer original insights into energy sustainability scenarios and provide valuable recommendations for policymakers, industry stakeholders, and researchers to identify the best ESSs according to the 3E aspects and improve their performance.

Furthermore, this study aims to enhance the understanding of integrating RESs and energy storage systems into energy supply chains, leading to more resilient and sustainable energy solutions.

2. Materials and methods

2.1. Reference building

In this work, a three-story residential building located in Tangier, Morocco, and connected to the electrical grid is considered as a base case. The primary features of this building are depicted in Table 1, while Fig. 1 provides data on direct normal irradiance, dry bulb temperature and wind speed for Tangier, characterized by a Mediterranean climate. Moreover, Table 2 illustrates the principal characteristics of energy systems taken into account in the reference ESS, designed to provide hot water, maintain a comfortable indoor environment, and power household electrical equipment. A heat pump with direct expansion coils is utilized for space heating and cooling, whilst an electric stratified water heater with a hot water storage tank is employed to provide hot water. These energy systems were meticulously chosen through rigorous optimization, aligning their capabilities with the specific requirements of the reference building. Parameters such as maximum flow rate of heating/cooling supply fan, nominal heating/cooling capacity of the heat pump, hot water storage tank volume, and coil nominal capacity were optimized using EnergyPlus software [22]. Additionally, consultations with the local market were conducted to select the other factors such as coil efficiency, heat pump coefficient of performance, and water heater efficiency. These data, along with others, have been taken into account within EnergyPlus to calculate the energy requirements of the reference building for heating, ventilation and air conditioning (HVAC), and providing hot water.

The results of modeling the energy needs of the reference building for water heating and air conditioning are depicted in Fig. 2. As this figure exhibits, the electricity demand for water heating is more significant than that for space heating and cooling. In addition, it is obvious that the energy demand of the HVAC system is more significant in summer than that in winter. This is mainly due to the Mediterranean climate of Tangier, which is distinguished by a predominance of cooling over heating [23]. Regarding the other household electrical equipment, their daily energy consumption on each floor is estimated at 5.57 kWh/day. These data are essential for sizing the energy systems in each ESS.

2.2. Proposed energy supply scenarios

Given the elevated energy demands of the reference building and the favorable climatic conditions (Fig. 1) for PV and WT installations, coupled with their exceptional cost-effectiveness in comparison with other RESs [24], assessing the 3E efficiency of PV and WT systems in

Table 1
Primary features of the explored building.

Feature	Value	Unit
Total floor area	396	m ²
Number of floors	3	–
Ceiling height	2.8	m
Building volume	1108.8	m ³
Number of facades	3	–
Occupants in each floor	4	peoples
Heat transfer coefficient of exterior walls	0.67	W/(m ² .K)
Heat transfer coefficient of interior walls	1.75	W/(m ² .K)
Heat transfer coefficient of floor	2.24	W/(m ² .K)
Heat transfer coefficient of ceiling	0.62	W/(m ² .K)
Heat transfer coefficient of exterior windows	3.3	W/(m ² .K)
Gross window-to-wall ratio	12.34	%
Air change per hour	0.5	volume/h
Cooling/heating setpoint	26/20	°C
Water consumption	102.28	m ³ /year

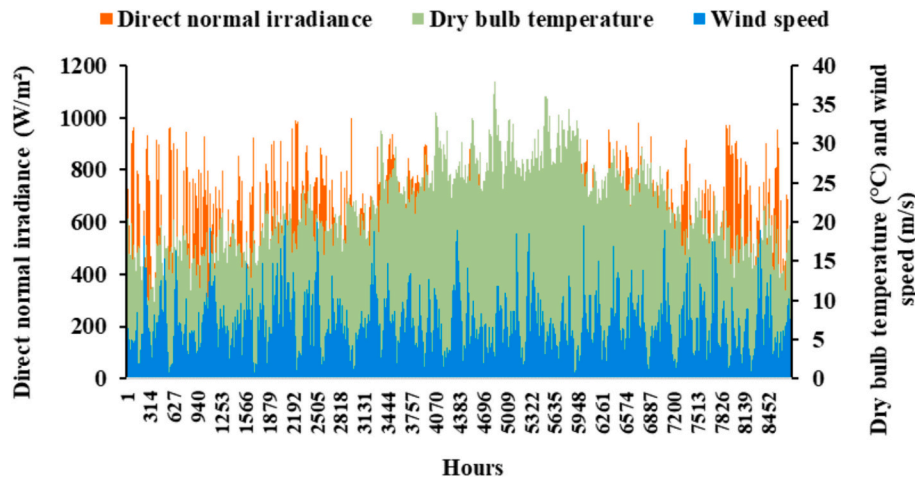


Fig. 1. Hourly weather data in Tangier city.

Table 2

Key features of the energy systems considered for each floor of the reference building.

Purpose	Device	Parameter	Input value	Unit
Space heating and cooling	Supply heating/cooling fan	Maximum flow rate	0.84	m ³ /s
		Total efficiency	80	%
	Heat pump	Nominal heating/cooling capacity	18.19	kW
		Gross rated sensible heat ratio	74.81	%
Water heating	Direct expansion cooling coil	Efficiency	90	%
	Electric heating coil	Nominal capacity	37	kW
	Stratified water heater	Maximum capacity	4.5	kW
		Maximum temperature limit	82.22	°C
	Hot water storage tank	Thermal efficiency	98	%
		Maximum hot water flow rate	3.36	L/min
		Volume	0.1893	m ³
		Height	1.4	m
		Shape	Vertical cylinder	–

supplying the building with clean electricity is of great importance. Additionally, there is a growing interest in evaluating the effectiveness of energy storage methods, mainly batteries and hydrogen storage, for storing excess electricity for later use. Therefore, this research examines

the 3E performance of three ESSs compared to the reference ESS where neither RES nor energy storage system is implemented and the required energy is supplied entirely from the grid. The overall schematic diagram of these three ESSs is exhibited in Fig. 3.

In the three ESSs, the required electricity to power electrical equipment, achieve comfortable air conditions and provide the building with hot water is planned to be supplied from the grid, the PV and WT systems, and/or the energy storage banks. The PV system of a total capacity of 13.2 kWp is configured to be installed on the entire building roof with a slope angle of 31° oriented towards south. These configurations (slope angle and orientation) have been selected based on their optimal performance identified in Ref. [25] under the climatic conditions of Tangier. The principal properties of this system are listed in Table 3. The second technology for producing electricity is the implementation of a horizontal axis WT (Table 3), which is installed to produce the remaining electricity demanded by the building. It should be emphasized that the investigated PV and WT systems are configured to maximize the direct supply of electricity to the building, while the surplus energy can be stored in two different devices: a battery unit and/or a hydrogen storage tank. For the battery bank, Li-ion batteries are used, owing to their elevated efficiency, extended lifespan, and compact physical dimensions relative to their capacity [26,27]. Regarding the electrolyzer and fuel cells, the proton exchange membrane (PEM) technology is considered owing to its maturity, substantial hydrogen production rate, elevated hydrogen purity and compact design [28,29]. Besides, the compressed hydrogen gas storage method is chosen because

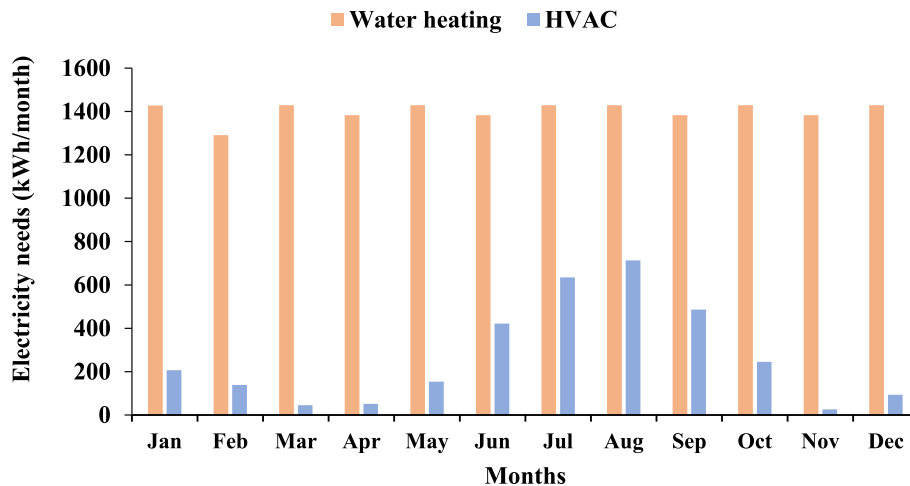


Fig. 2. Electricity needs for water heating and HVAC systems in the reference building.

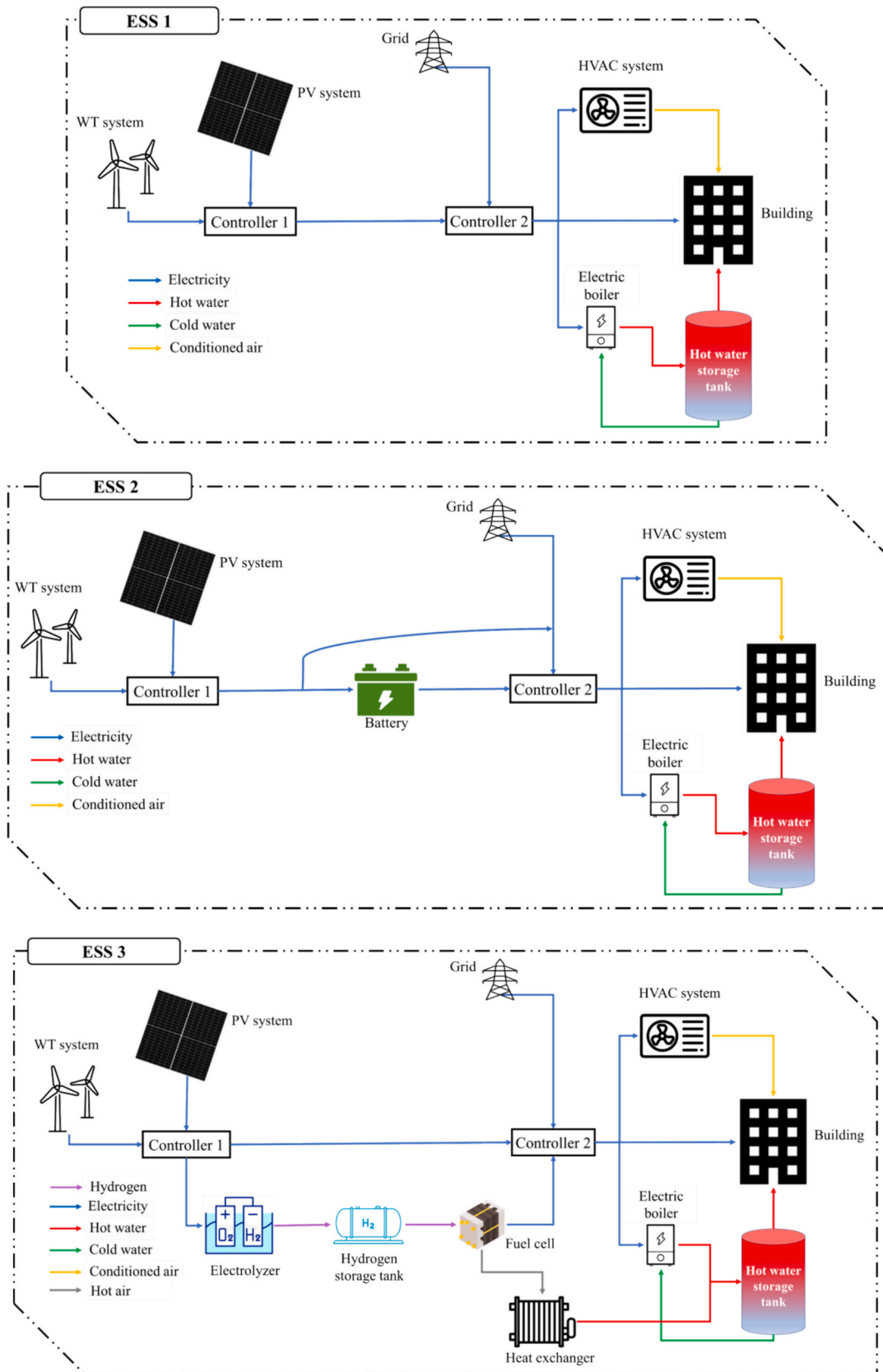


Fig. 3. Schematic diagram of the proposed ESSs.

Table 3
Primary properties of the studied systems.

Feature	Value/type	Unit
PV system		
Number of PV panels (series × parallel)	24 (12 × 2)	–
Panel dimensions	2278 × 1134 × 35	mm
Maximum efficiency of PV panels	21.32	%
Array slope	31	°
Array azimuth	0	°
Nominal power of the inverter	10	kW
Maximum PV power allowed by the inverter	15	kW
Maximum efficiency of the inverter	97.8	%
WT system		
Rated rotor speed	200	rev/min
Rotor diameter	6.3	m
Number of blades	3	–
Rated power	7.5	kW
Rated wind speed	12	m/s
Cut-in wind speed	3	m/s
Cut-out wind speed	20	m/s
Annual local wind speed	4.9	m/s
Hydrogen systems		
Electrolyzer type	PEM	–
Electrolyzer mean electrical conversion efficiency	70	%
Electrolyzer rated capacity	14	kW
Hydrogen storage type	Compressed gas	–
Hydrogen storage tank capacity	2.7	kg
Density of the stored hydrogen	7.7	kg/m ³
Lower heating value of hydrogen	33.33	kWh/kg
Fuel cell type	PEM	–
Fuel cell mean electrical conversion efficiency	40	%
Fuel cell rated capacity	14	kW
Heat exchanger capacity	10	kW
Battery system		
Battery type	Li-ion	–
Rated capacity of the battery unit	45	kWh
Charging/discharging efficiency of batteries	90	%

of its relatively high energy density, maturity, proven safety and cost-effectiveness [30].

3. Modeling and evaluation procedures

3.1. Modeling of energy supply scenarios

To model the investigated ESSs, multiple software tools have been employed. Fig. 4 displays the main stages of simulating the proposed ESSs. First, to calculate the energy needs of the reference building, the latter was designed utilizing SketchUp software [31] and the OpenStudio plugin [32]. The established model was then exported as an EnergyPlus file to input other properties of the reference building, such as envelope materials, internal gains, and HVAC system features. Regarding electrical needs for water heating, the WaterHeater:Stratified object was used to specify the main characteristics of the utilized water heater and hot water storage tank. The energy balance of a hot water tank, used in EnergyPlus, can be expressed as indicated by Eq. (1) [33]. EnergyPlus software was used due to its comprehensive modeling capabilities, high accuracy and extensive validation in building energy simulation [34–37].

$$m c_p \frac{dT_{ST}}{dt} = Q_{boiler} - Q_{loss} - Q_{load} \quad (1)$$

in which m is the total mass of water inside the tank. c_p is the specific heat of water, while T and t are the water temperature in the tank and time, sequentially. The thermal energy delivered by the electrical boiler is denoted as Q_{boiler} , while Q_{loss} represents the rate of energy loss from the storage tank to the surrounding environment. Finally, Q_{load} is the amount of energy extracted from the tank to meet the hot water needs.

On the other side, to estimate the electrical output of the considered WT system, the Generator:WindTurbine object within EnergyPlus was applied, in which numerous parameters like those listed in Table 3 were entered. This object employs a simple approximation approach to determine the coefficient of performance (C_p) and actual power output of a WT (P_{WT}), including conversion and supply losses. The coefficient of performance and capacity factor (CF) of a WT can be calculated using Eqs. (2) and (3) [38]. Concerning the PV system's energy production, it was estimated using PVsyst software [39]. The energy yields (E_{PV}) and the performance ratio (PR) of the examined PV system, revealed by PVsyst, can be obtained using Eqs. (4–6). This software was employed

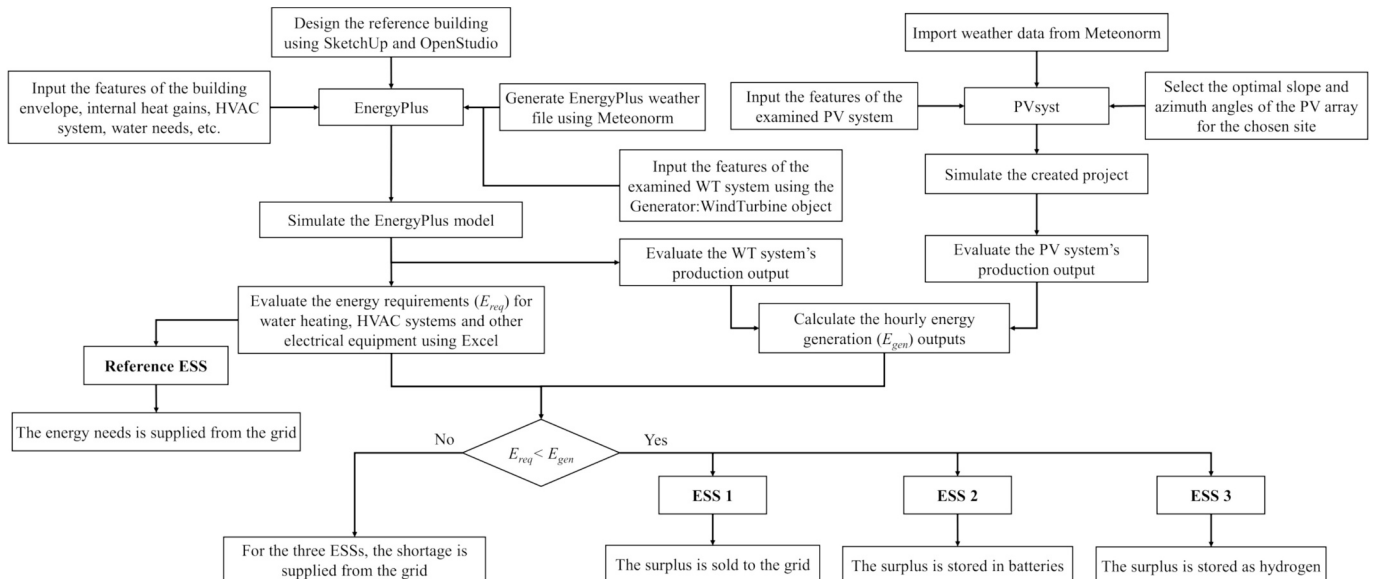


Fig. 4. Flowchart of the primary steps to simulate the studied ESSs.

because of its detailed simulation capabilities, extensive database of PV components, and proven accuracy in predicting system performance, resulting in reliable and optimized design results [40].

$$C_p = \frac{P_{WT}}{0.5\rho A_r V^3} \quad (2)$$

$$CF = \frac{E_{WT}}{P_w \times h} \times 100 \quad (3)$$

$$E_{PV} = A_{PV} \times f_{active} \times G_T \times \eta_{cell} \times \eta_{invert} \quad (4)$$

$$\eta_{cell} = \eta_{STC} [1 - \beta(T_{cell} - T_{STC})] \quad (5)$$

$$PR = \frac{E_{PV}}{E_{PV-STC}} \quad (6)$$

where P_w denotes the rated power output at the rated wind speed V , and E_{WT} expresses the amount of energy generated by the WT in a specific period. ρ and A_r signify the air density and swept area of rotor, respectively. h indicates the number of hours in a given time frame (e.g., a month or year). A_{PV} is the total surface area of the PV module, while f_{active} is its fraction with active PV cells. G_T describes the overall solar radiation reached the PV module, whereas η_{cell} and η_{invert} are the electrical efficiency of the solar cells and inverter, respectively. E_{PV-STC} and η_{STC} correspond to the ideal PV system's energy output and the nominal electrical efficiency of the PV panel measured at standard test conditions of 1000 W/m² irradiance and 25 °C cell temperature (T_{STC}). T_{cell} and β represent the cell temperature and temperature coefficient, consecutively.

To evaluate both investigated storage techniques, a mathematical model is used to calculate the amount of electricity produced from generators (E_{gen}) and comparing it to the requested electrical loads (E_{req}). Any surplus electricity is stored for future use, and in case of a shortfall, the remaining electricity demand is drawn from storage systems. The energy charged to/discharged from (E_{ch}/E_{disch}) a storage system can be computed using Eqs. (7) and (8).

If $E_{req} < E_{gen}$:

$$E_{ch} = E_{gen} - E_{req} \quad (7)$$

Else:

$$E_{disch} = E_{req} - E_{gen} \quad (8)$$

For hydrogen storage and consumption calculations, Eqs. (9–11) are used to estimate the consumed power by the incorporated electrolyzer (E_{ele}), the amount of hydrogen produced (m_{H_2}), along with the fuel cells' power (E_{FC}) and thermal energy (Q_{FC}) outputs [15,41–43]. Additionally, the quantity of electricity stored in the battery unit during charging and discharging states are calculated using Eqs. (12) and (13), respectively, while its autonomy (B_{aut}) is computed utilizing Eq. (14) [44].

$$E_{ele} = \frac{m_{H_2} \times LHV_{H_2}}{\eta_{ele}} \quad (9)$$

$$E_{FC} = m_{H_2} \times LHV_{H_2} \times \eta_{FC} \quad (10)$$

$$Q_{FC} = E_{FC} \left(\frac{1 - \eta_{FC}}{\eta_{FC}} \right) \quad (11)$$

$$E_{stor}^{t+\Delta t} = E_{stor}^t + E_{ch} \times \eta_{B-ch} \times \Delta t \quad (12)$$

$$E_{stor}^{t+\Delta t} = E_{stor}^t - \frac{E_{disch} \times \Delta t}{\eta_{B-disch}} \quad (13)$$

$$B_{aut} = \frac{V_B \times C_B \times \left(\frac{100 - SoC_{min}}{100} \right)}{P_B} \quad (14)$$

where LHV_{H_2} denotes the hydrogen's lower heating value. η_{ele}/η_{FC} is the electrolyzer/fuel cell efficiency, while $\eta_{B-ch}/\eta_{B-disch}$ designates the charging/discharging efficiency of the battery system. $E_{stor}^{t+\Delta t}$ signifies the quantity of electricity stored in the battery system at the time of charging or discharging processes, while E_{stor}^t is the stored electricity when neither process is taking place. Δt is the time step and V_B/C_B is the nominal voltage/capacity of a single battery. SoC_{min} represents the minimum state of charge and P_B stands for the load power of the battery.

3.2. Evaluation criteria

This section presents the primary metrics used to assess the effectiveness of the studied ESSs. In energy terms, the performance of these scenarios is evaluated employing the parameters outlined in Eqs. (1)–(14) and the energy coverage ratio (ECR). The latter is an essential indicator as it indicates the proportion of energy demand met by the integrated RESs, hence evaluating the feasibility of ESSs by considering the balance between renewable energy generation and demand. This metric can be calculated using Eqs. (15)–(17) [45].

$$ECR_1 = \frac{E_{PV} + E_{WT}}{E_{req}} \quad (15)$$

$$ECR_2 = \frac{E_{PV} + E_{WT} + E_B}{E_{req}} \quad (16)$$

$$ECR_3 = \frac{E_{PV} + E_{WT} + E_{FC}}{E_{req}} \quad (17)$$

where ECR_1 , ECR_2 and ECR_3 represent the ECR for ESS 1, 2 and 3, respectively. E_{PV} , E_{WT} , E_B and E_{FC} are the energy provided by PV, WT, battery unit and fuel cells, sequentially.

To assess the environmental friendliness of the explored systems according to every ESS, two primary indicators are used: CPBP and CER. They were chosen for their ability to thoroughly assess the environmental impact and sustainability of different ESSs. The CPBP is employed to estimate the period that a system will take to offset the emissions generated during its manufacturing, implementation and operation, highlighting its long-term environmental benefits. Besides, the CER is used to calculate the amount of CO₂ emission reductions utilizing the examined ESSs compared to the reference scenario in which all of the reference building's needs are supplied by the local grid, with a carbon emission factor (CEF) of 0.547 kgCO_{2e}/kWh [46]. These metrics are evaluated using Eqs. (18) and (19) [47,48]. Indeed, evaluating these indicators offers valuable information for decision-makers to identify the ESSs with minimal environmental impacts.

$$CPBP = \frac{EC}{C_S} \quad (18)$$

$$CER = E_{SC} \times CEF - EC \quad (19)$$

where EC is the embodied carbon of the examined system, while C_S is its resulted annual carbon savings. The embodied carbon values of the main systems used in this paper have been meticulously taken from the literature and are summarized in Table 4. E_{SC} expresses the overall energy supplied to the building and/or injected into the grid using each investigated ESS.

Table 4
Embodied carbon of the main systems explored in this study.

Element	Embodied carbon	Unit	Ref.
PV system	267.5	kgCO _{2e} /m ²	[49]
WT	11	gCO _{2e} /kWh	[50]
Battery	200	kgCO _{2e} /kWh	[51]
Hydrogen	1.26	gCO _{2e} /gH ₂	[8]

The economic profitability of the explored ESSs is assessed utilizing five distinct metrics, specifically SPP, DPP, SIR, LCOE and LCOH. These metrics were selected because of their comprehensive effectiveness in evaluating the economic viability of ESSs. The SPP specifies the time required to recover the initial investment in a project. DPP is quite similar to SPP with the slight difference that DPP considers the time value of money, indicating the long-term financial attractiveness of the project. SIR compares the savings to the upfront investment to help evaluate the profitability of a given system, and the LCOE/LCOH estimates the cost of energy/hydrogen production. The formulas for calculating these metrics are given in Eqs. (20)–(24) [40,52]. Assessing these indicators will provide a comprehensive economic analysis, helping stakeholders make appropriate decisions, examine long-term viability, and reduce financial risks associated with each studied ESS.

$$SPP = \frac{CAPEX}{C_F} \quad (20)$$

$$DPP = \frac{\ln\left(\frac{1}{1 - \frac{CAPEX}{C_F d}}\right)}{\ln(1 + d)} \quad (21)$$

$$SIR = \frac{\sum_{t=0}^N \frac{S_t}{(1+d)^t}}{\sum_{t=0}^N \frac{I_t}{(1+d)^t}} \quad (22)$$

$$LCOE = \frac{CAPEX + OPEX}{\sum_{t=1}^N \frac{E_t}{(1+d)^t}} \quad (23)$$

$$LCOH = \frac{CAPEX + OPEX}{\sum_{t=1}^N \frac{(H_2)_t}{(1+d)^t}} \quad (24)$$

in which *OPEX* represents the operating and maintenance costs. Table 5 outlines the main costs taken in the economic evaluation. C_F refers to the annual cash flow, while S_t , I_t , E_t and $(H_2)_t$ express the savings, investments, the generated/stored energy and the amount of hydrogen produced during the *t*-th year, respectively. The system's lifetime is denoted by *N*, and the discount rate is represented by *d*.

3.3. Model validation

In this research, to meticulously calculate the energy needs for both HVAC and water heating systems in the reference scenario, alongside assessing the energy yields from the examined WT, EnergyPlus software

was utilized. The accuracy of this tool in evaluating the space heating and cooling loads, as well as energy requirements for water heating, has been verified in many studies in the literature [59–61]. In addition, the reliability of the WT generator model integrated in EnergyPlus has been confirmed by numerous researchers [62]. This collective body of research strengthens the robustness and reliability of the instruments used in the current study, providing a solid foundation for the obtained energy-related calculations and analyses. Similarly, the accuracy of PVsyst software, utilized in this research to calculate the energy outputs of the investigated PV system, has been validated by several studies [63–65]. To further corroborate the findings of the explored PV system, the productibility and the performance ratio of the investigated PV system are compared with those found in the experimental study conducted by Ameer et al. [66] under comparable climatic conditions. The values obtained in the present study are 5.44 Wh/Wp/day and 83.92 %, while Ref. [66] found values of up to 5.23 Wh/Wp/day and 86.3 %, with corresponding relative errors of 4 % and 2.76 %, respectively.

Concerning the hydrogen calculation method applied in this paper, it is validated by comparing the LCOH achieved in this study with that acquired by Berrada and Laasmi [67]. It should be noted that this previous work used a similar methodology to evaluate the outputs of the hydrogen systems used in the current paper, and their findings were validated by experimental and analytical studies published in the literature. In the specific context of supplying electricity to PEM electrolyzer from a PV system in the Mediterranean climate of Morocco, the resulting LCOH in Ref. [67] was 5.8 \$/kg, whilst in our study it is approximately 5.3 \$/kg. Comparing the LCOH in these two studies yields a relative error of 8.6 %.

4. Results and discussion

4.1. Energy performance

In this section, the energy performance of the investigated ESSs is analyzed. Fig. 5 exhibits the amount of power generation using the examined PV and WT systems. From this figure, it can be seen that a large amount of electricity is generated using the studied PV and WT systems, with an annual energy production of 26,238.44 and 8432.49 kWh, respectively. These values correspond to a productivity of 1988 Wh/Wp/year for PV system and an annual energy generation of 1124.33 kWh per unit of rated power of WT. Actually, the maximum share of hourly power production is 12.11 kW for the PV system and 6.26 kW for the WT system. These values are obtained during the periods when the PV system receives the highest solar radiation and when the WT site records the highest wind speed. These findings are essential for comparing the energy performance of the two investigated RESs and therefore assessing their environmental and economic effectiveness.

The energy requirements of the examined building detailed in Section 2.1 and the power outputs data of the studied PV and WT installations presented in Fig. 5 are used to estimate the overall energy

Table 5

Main economic data considered for all ESSs.

	Cost	Unit	Considered lifetime, years	Ref.
CAPEX				
PV system	800	\$/kW	25	Local market
Battery	150	\$/kWh	10	[45]
WT	1325	\$/kW	25	[24]
Electrolyzer	1200	\$/kW	20	[53]
Hydrogen tank	378	\$/kgH ₂	25	[54]
Fuel cells	1000	\$/kW	10	[55]
Heat exchanger	500	\$	15	Local market
Discount rate	3	%	–	[56]
Electricity price	0.116	\$/kWh	–	[57]
Annual OPEX	2	% of CAPEX	–	[58]

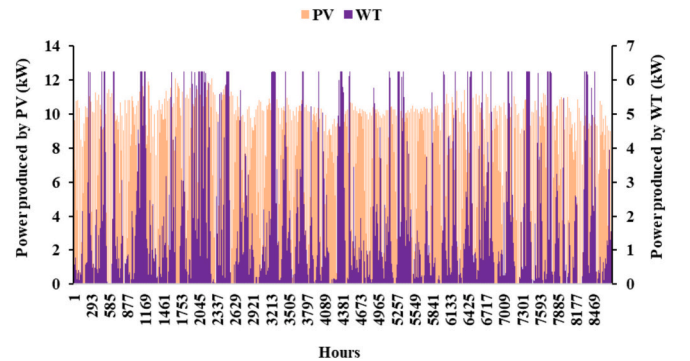


Fig. 5. Power production using the considered PV and WT systems.

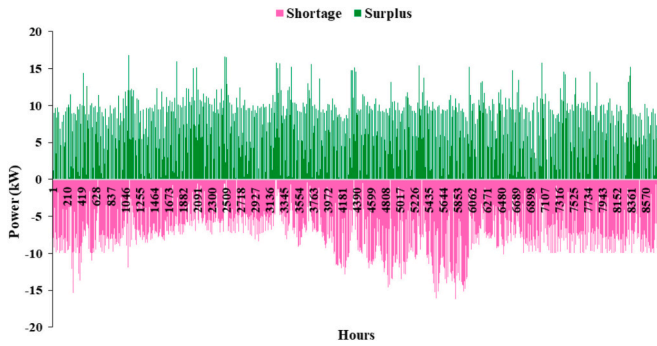


Fig. 6. Power surplus and shortage in the first ESS.

surplus and shortage in the three studied ESSs. The findings of this evaluation for the first ESS are outlined in Fig. 6. This figure highlights that the annual energy surplus (20,935.10 kWh) is 68.83 % higher than energy shortage (12,399.86 kWh). Furthermore, it can be seen that the energy shortage increases significantly during the summer period (i.e., 4425th–6324th hours), which can mainly be attributed to the elevated cooling demand in this period of the year (see Fig. 2).

The above data are exploited to analyze the energy outcomes of the second and third ESSs. The primary energy findings of these scenarios are presented in Figs. 7 and 8, consecutively. Fig. 7 shows that delivering the excess electricity produced by the considered PV and WT systems to the batteries (i.e., in the second ESS) is an energy-efficient method, since it provides all the energy requirements of the building under study. In this scenario, the overall electricity charged to/discharged from batteries is about 18,842/16,957 kWh/year. On the other hand, Fig. 8 reveals that supplying the excess electricity of PV and WT generators, depicted in Fig. 7, to the selected electrolyzer (i.e., in the third ESS) produces around 440 kg/year of hydrogen, corresponding to 14,655 kWh/year of energy. Furthermore, the findings point out that consuming the generated hydrogen via the considered fuel cells results in annual electricity and heat production of 5862 and 8793 kWh/year, respectively.

To better compare the energy outputs of the three investigated ESSs, these outputs are summarized in Fig. 9. This figure emphasizes that the second and third ESSs provide the highest ECR with a value of 100 %, while the first ESS achieves lower ECR of 52.56 %. This highlights the importance of incorporating both RESs and energy storage systems, mainly batteries and hydrogen, into the energy supply chain to achieve energy independence, thereby reaching the goal of net zero energy buildings. In contrast, the total amount of energy fed into the grid is more significant in the first ESS, where no storage system is integrated, compared to the second and third ESSs, with values of 20,935, 4558 and 2255 kWh/year, respectively. These results can be explained by the minimum losses of the first ESS, and the increased energy efficiency of

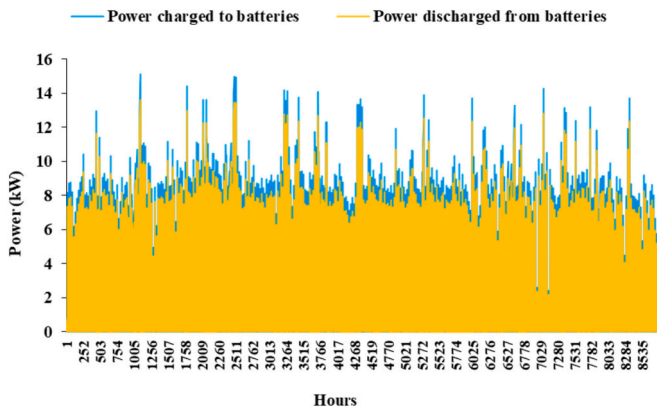


Fig. 7. Power charged to/discharged from batteries in the second ESS.

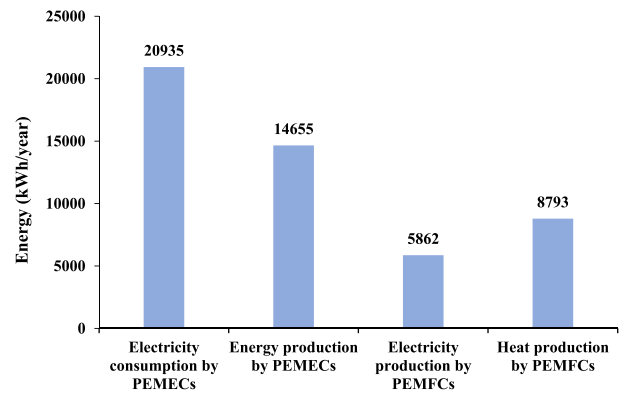


Fig. 8. Main energy outcomes of the third ESS.

batteries compared to the hydrogen systems where significant losses occurred. Therefore, it is vital to diminish the energy losses within hydrogen supply chain, with a specific focus on optimizing efficiency in electrolyzers and fuel cells. This approach is critical not only to enhance the overall energy efficiency of hydrogen use, but also to merge it seamlessly into the energy distribution network.

4.2. Environmental analysis

The energy findings reported above and the embodied carbon of the materials integrated in every system, listed in Table 4, are used to estimate the CPBP and CER of the explored ESSs. The results of this assessment are shown in Fig. 10. This figure illustrates that the first ESS, relying solely on the direct energy supply without using any energy storage method, yields the most favorable environmental outcomes with the shortest CPBP (one year) and the greatest CER (18.21 tCO_{2e}/year). This result can be interpreted by the significant amount of green energy supplied to the building and fed into the grid, which is much higher in the first ESS than in the two others (see Fig. 9). Additionally, the first ESS includes fewer materials than the second and third ESSs, which encompass additional equipment for energy storage, thereby raising their carbon footprint. On the other hand, it is noticeable that the second ESS exhibits longer CPBP but slightly higher CER values compared to the third ESS, with values of 1.66 years and 15.13 tCO_{2e}/year for the second ESS, and 1.14 years and 14.82 tCO_{2e}/year for the third ESS, respectively. These outcomes can be ascribed to the elevated embodied carbon of batteries compared to the hydrogen storage method, resulting in an extended CPBP for the second ESS in comparison with the third one. This point has been confirmed in several previous works, such as in Refs. [68–70]. Furthermore, the increased quantity of annual energy supplied to the building in the second ESS, as shown in Fig. 9, elucidates the slight relative rise in its CER compared to the third ESS. In real-world settings, these environmental outcomes could guide decisions regarding the selection of environmentally friendly ESSs, thereby reducing the carbon footprint of the energy sector and improving building sustainability.

From the above findings, it is clear that the incorporation of RESs for energy supply has a substantial role in diminishing the carbon emissions of buildings. However, the embodied carbon of some energy storage equipment, such as batteries and hydrogen storage systems, hinders the environmental feasibility of these technologies. Therefore, minimizing the carbon content of equipment used in the energy supply infrastructure as well as providing affordable and efficient recycling techniques are essential to improving the environmental friendliness of energy supply routes using storage methods, like in the second and third ESSs examined in this research.

4.3. Economic feasibility

In this section, the economic viability of the studied ESSs is explored,

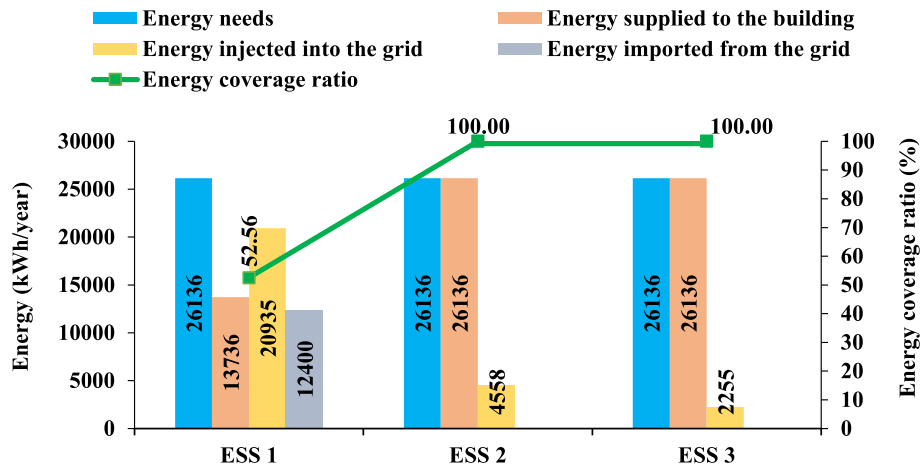


Fig. 9. Main energy outcomes for every ESS.

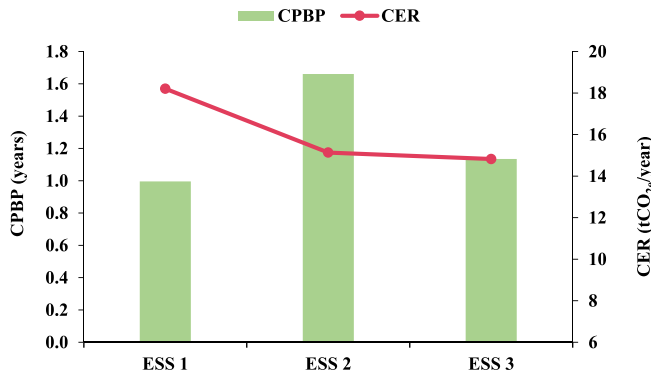


Fig. 10. CPBP and annual CER for each investigated ESS.

and the primary outcomes are shown in Fig. 11. As this figure emphasizes, the first ESS reveals the best economic findings with the shortest SPP/DPP, the lowest LCOE and the highest SIR. These shares are around 5.80/6.46 years, 0.05 \$/kWh and 2.26, respectively. These significant economic outcomes of the first ESS can be attributed to the large quantity of energy generated by the PV and WT systems, which is designed to be directly supplied to the considered building or injected

into the grid. This study assumes that the price of electricity sold to the grid is 80 % of that purchased from it [47]. In addition, the results highlight that the second ESS, which employs batteries to store electricity, is more economically feasible than the third ESS that uses hydrogen for energy storage. This conclusion is supported by the respective values of SPP, DPP, LCOE and SIR, which amount to 7.89 years, 9.13 years, 0.09 \$/kWh and 1.44 for the second ESS, while the third one exhibits the values of 16.30 years, 22.71 years, 0.21 \$/kWh and 0.66. The higher cost-effectiveness of the second ESS compared to the third one is mainly due to the smaller capital costs required to incorporate batteries in contrast to hydrogen systems, which are characterized by their substantial CAPEXs (Table 5). The minimal losses encountered when using batteries are also a critical factor that increases the profitability of this energy storage technology compared to the hydrogen storage one that still suffers from many energy losses. These findings are in line with those reported by Koholé et al. [71], highlighting that storing surplus energy generated by PV and WT systems in batteries rather than as hydrogen proved to be a more cost-effective choice. From another point of view, Zhang et al. [72] indicated that the electrolyzer cost is the most critical factor to enhance the economic viability of hydrogen storage. Furthermore, Falama et al. [73] outlined that enhancing fuel cell energy efficiency is essential for lowering the cost of hydrogen-based energy storage systems. This would certainly increase the competitiveness of hydrogen as a means of energy storage.

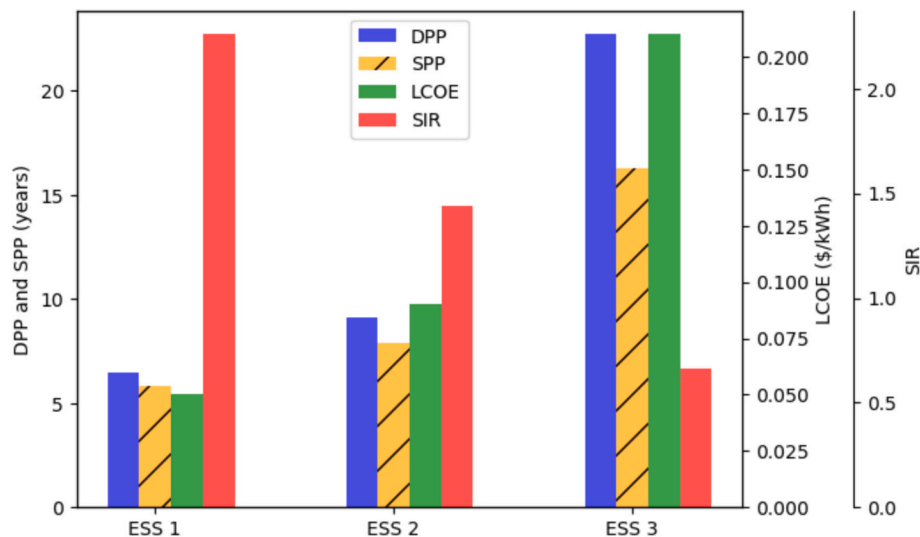


Fig. 11. Economic indicators of each studied ESS.

To summarize, the above results clearly underline that direct supply of renewable energy and the use of batteries for electricity storage provide significant economic benefits. On the contrary, the inclusion of hydrogen for energy storage is not economically viable. In this regard, improving the efficiency of hydrogen systems and launching support programs to reduce their costs, mainly electrolyzers, hydrogen storage tank and fuel cells, is essential for enhancing the cost-effectiveness of the third ESS. Besides, as demonstrated in Ref. [68], the introduction of carbon pricing policies could have a significant effect on augmenting the economic benefits of energy storage in the form of hydrogen, thereby encouraging households to adopt such ESSs.

4.4. Limitations of this study

First, it is important to note that this study was based on simulation data, which may introduce uncertainties in real-world settings regarding some parameters, such as actual weather conditions, maintenance costs, and embodied carbon. Second, while the economic analysis of ESSs offered insights into their cost-effectiveness, it did not account for potential future changes in energy prices or policy incentives that could impact the economic viability of these ESSs. In addition, this study focused on three of the most efficient and promising ESSs. However, other ESSs have also demonstrated high efficiency, underscoring the importance of comparing their effectiveness with those examined in this research. This comparison will only be conclusive if other ESSs are assessed under similar climatic, economic, and environmental circumstances as those considered in this study.

Indeed, addressing the above limitations in future works would enhance the robustness and applicability of the study's findings. Nonetheless, it is worth mentioning that we have minimized the uncertainty of the results as much as possible by incorporating realistic parameters, such as building and HVAC system features (Section 2.1), embodied carbon (Table 4), and the CAPEX and OPEX of all scenarios (Table 5). These parameters were carefully selected based on a thorough review of local thermal regulations, the local market, and consultations with local companies in the field.

5. Conclusions

In this research, the energy, environmental and economic (3E) performance of three energy supply scenarios was assessed. The explored scenarios were designed to provide electricity, heating and cooling to a typical residential building under the Mediterranean climate of Tangier, Morocco. The first scenario was intended to directly feed the energy, generated by the examined photovoltaic and wind turbine systems, into the building and inject the extra electricity into the grid, while the second and third scenarios involved storing this excess energy in batteries and as hydrogen, respectively. To achieve this aim, multiple software tools, mainly EnergyPlus and PVsyst, and numerical models were used. Furthermore, the 3E efficiency of the three scenarios was analyzed utilizing key metrics, including ECR, CPBP, CER, SPP, DPP and LCOE. The major conclusions from this study can be drawn as follows.

- i. The second and third scenarios achieved the highest ECR with a value of 100 %, whilst the first scenario showed lower ECR of 53.56 %. In contrast, the first and second scenarios exhibited the greatest energy efficiencies, delivering more energy than the third scenario.
- ii. Environmentally, the first scenario revealed the excellent outcomes with a CPBP of one year and a CER of 18.21 tCO_{2e}/year. In addition, the third scenario outperformed the second one in terms of environmental friendliness, as the former reached a lower CPBP of 1.14 years compared to 1.66 years for the latter.
- iii. From an economic perspective, the first scenario showed the best economic findings followed by the second one, while the third scenario exhibited unfavorable economic results. The obtained

SPP/DPP values for the first, second and third scenarios were found to be 5.80/6.46, 7.89/9.13 and 16.30/22.71 years, whereas their LCOE were 0.05, 0.09 and 0.21 \$/kWh, sequentially. Indeed, the findings of this research could be relevant in various regions worldwide, characterized by similar conditions in terms of weather data, environmental parameters (e.g., carbon emission factors), and economic variables including energy prices, discount rates, and economic incentives.

In summary, the outcomes of this work highlighted the substantial 3E effectiveness of the first and second scenarios; however, the third scenario, wherein energy is stored in the form of hydrogen, still faces many challenges, especially with regard to energy and economic prospects. Therefore, more research and development is necessary to enhance the energy performance and cost-effectiveness of hydrogen production, storage and utilization systems, leading to increase their widespread adoption and improve building sustainability.

CRedit authorship contribution statement

Y. Elaouzy: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **A. El Fadar:** Writing – review & editing, Validation, Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **O.B. Achkari:** Writing – review & editing, Investigation, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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